LingLab at ArGuard Shared Tasks: A Lightweight Hybrid Classifier for Arabic Harmful Prompt Detection

O. O. Adebanji[2], E. O. Ilesanmi[1], O. E. Ojo[2], H. Calvo[2], G. Sidorov[2], A. Feldman[3]

[1]Federal University Oye-Ekiti, Nigeria

[2]Instituto Politécnico Nacional, Center for Computing Research, Mexico City, Mexico

[3]Montclair State University, Montclair, NJ, USA

{olaronke.oluwayemisi, emmanuelilesanmi43, olumideoea}@gmail.com

{hcalvo, sidorov}@cic.ipn.mx, feldmana@montclair.edu

Abstract

The classification of Arabic prompts as *Safe* or *Unsafe* is the focus of this submission to ArGuard 2026 Subtask B1. We developed a lightweight hybrid classifier that combines lexical and semantic information. The lexical branch uses normalized word-level and character-level features with a class-balanced LinearSVC classifier. The semantic branch uses frozen multilingual E5 embeddings with a class-balanced logistic regression classifier. We combined the outputs of both branches using late fusion. On the development set, LinearSVC achieved a Macro-F1 score of 0.93, while the frozen E5 branch achieved 0.88. The final hybrid system achieved a Macro-F1 score of 0.93. On the test set, the submitted system achieved 0.78 accuracy, 0.83 precision, 0.70 recall, and 0.72 Macro-F1.

1 Introduction

Classification systems assign predefined labels to input data based on patterns learned from labeled examples (Adebanji et al., 2022, 2024; Ojo et al., 2020, 2023, 2024). In this work, the task is to classify Arabic prompts as either *Safe* or *Unsafe*. Such classifiers can support content-safety pipelines by identifying potentially harmful prompts before they are processed by a generative system. However, safe and unsafe prompt classification is not a simple keyword-detection task. Many safe prompts contain words related to violence, fraud, self-harm, harassment, hate, or illegal activities. For example, users may ask for educational explanations, prevention advice, reporting procedures, medical support, or information about the consequences of harmful actions. A reliable classifier must therefore distinguish harmful intent from legitimate discussion of sensitive topics.

The task is especially challenging for Arabic. Arabic includes regional dialects, informal writing styles, spelling variation, code-switching, transliteration, and Arabizi. The same request can be written in several different forms depending on the speaker, location, platform, and writing style. These variations can reduce the performance of classifiers that depend only on exact words or standard Arabic spelling. Previous work has shown that Arabic safety evaluation should account for cultural context, dialectal variation, and different writing systems (Gupta et al., 2026; Bouke et al., 2026; Singh, 2026; Shanshin, 2026; Saud et al., 2026).

ArGuard 2026 addresses harmful-content detection in Arabic memes and prompts directed at LLMs (Alam et al., 2026; Kmainasi et al., 2026). Task B focuses on Arabic harmful-prompt detection. In Subtask B1, each prompt must be classified as either *Safe* or *Unsafe* (Zaghouani et al., 2026). The task requires systems that can identify unsafe requests while avoiding the incorrect classification of safe prompts that mention sensitive topics. Our system combines lexical and semantic information through a lightweight hybrid classifier to address this task. The lexical branch uses normalized word-level and character-level features with a LinearSVC classifier. Word features capture common expressions and short phrases, while character features help handle spelling variation, Arabic morphology, and informal text. The semantic branch uses multilingual-E5 embeddings (Wang et al., 2024). with logistic regression. This branch provides a broader representation of prompt meaning without requiring end-to-end transformer fine-tuning. We standardize the scores produced by both branches and combine them through a development-based late-fusion search.

Figure 1 shows the overall architecture of the proposed system. The system standardizes the output scores from both branches and combines them using weights of 0.80 for the lexical branch and 0.20 for the semantic branch. The main contributions of this work are a lightweight Arabic prompt-safety classifier based on word-level and character-level

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features, a frozen multilingual semantic branch that avoids expensive transformer fine-tuning, a score-standardized late-fusion approach that combines lexical and semantic predictions, and an analysis of the substantial gap between development and test performance.

2 Data, Preprocessing, and Resources

2.1 Task Data

After standardization, the training data contained 16,831 labeled prompts. We removed prompts with the same normalized text but different labels. We also removed training prompts that exactly matched prompts in the development set after normalization. The final training set used for model selection contained 16,776 prompts. The development set contained 2,405 prompts, and the blind test set contained 5,979 prompts. The data were imbalanced. Unsafe prompts represented 79.8% of the model-selection training set and 80.2% of the development set. Therefore, we used class-balanced classifiers.

2.2 Text Preprocessing

We normalized Arabic text to reduce spelling differences. URLs, user mentions, and hashtag symbols were replaced with `URL`, `USER`, and `HASHTAG`. We removed Arabic diacritics and tatweel. We also normalized alef variants, alef maqsura, hamza forms, repeated characters, Latin letters, and repeated spaces. For the lexical model, we included dialect information when it was available:

```
dialect <d> text <normalized prompt>
```

For the E5 model, we used the original prompt in the following format:

```
query: dialect: <d>. text: <original
prompt>
```

When dialect information was not available, we used only the prompt text.

3 System Architecture

Figure 1 presents the proposed hybrid classifier for Arabic harmful-prompt detection. The system receives an Arabic prompt together with its dialect label and processes the input through two complementary branches. The lexical branch operates on normalized, dialect-aware text, whereas the semantic branch encodes the original prompt using the E5 query format. The resulting branch scores are standardized and combined through late fusion to predict either *Safe* or *Unsafe*.

3.1 Lexical Branch

The lexical branch first normalizes the Arabic prompt and adds the available dialect information. It then extracts word- and character-level TF-IDF features. The word vectorizer uses 1 - 2 grams with a maximum of 50,000 features, `min_df=2`, `max_df=0.999`, and sublinear term-frequency scaling. The character-boundary vectorizer uses 3 - 5 grams with a maximum of 100,000 features, `min_df=2`, and `max_df=0.9995`. We trained a class-balanced `LinearSVC` classifier with $C = 1.25$, a maximum of 4,000 iterations, and random seed 42. The decision value associated with the *Unsafe* class was used as the lexical score. On the development set, this branch achieved a Macro-F1 of 0.9326 and an accuracy of 0.9576. Its class-specific F1 scores were 0.8915 for *Safe* prompts and 0.9736 for *Unsafe* prompts.

3.2 Semantic Branch

The semantic branch uses the original prompt and converts it to the E5 query format. When dialect information is available, the input is formatted as follows:

```
query: dialect: <d>. text: <original
prompt>
```

We used the frozen `intfloat/multilingual-e5-small` encoder to generate semantic representations. Its parameters were not fine-tuned on the task data, and each prompt was truncated to a maximum of 128 subword tokens. Given the resulting token-level hidden representations, we computed a single prompt embedding using attention-mask-aware mean pooling:

$$\mathbf{e} = \frac{\sum_t m_t \mathbf{h}_t}{\max\left(\sum_t m_t, \epsilon\right)}, \tag{1}$$

where $\mathbf{h}_t$ denotes the hidden representation of token t, m_t is its attention-mask value, and ϵ is a small constant that prevents division by zero. The resulting embedding was subsequently L2-normalized.

We trained a class-balanced logistic regression classifier on the fixed E5 embeddings. The regularization parameter was selected from $C \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$ using development Macro-F1. Among the evaluated values, $C = 4.0$ achieved the highest development score. Because this value was the upper boundary of the search range, we treat it as the best tested configuration

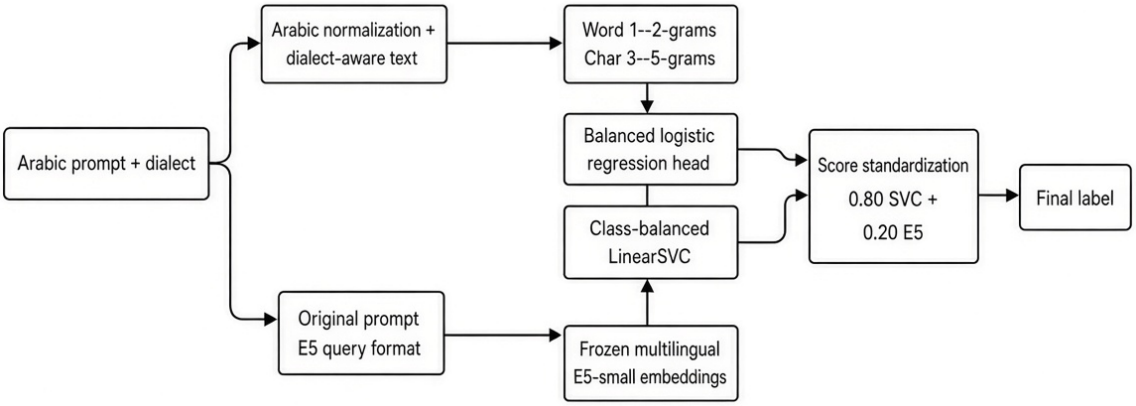


Figure 1: Architecture of the proposed hybrid Arabic prompt-safety classification system

rather than a demonstrated optimum; evaluating larger values would be necessary to determine whether performance had reached a plateau. With $C = 4.0$, the E5 branch achieved a development Macro-F1 of 0.8797 and an accuracy of 0.9222. Its class-specific F1 scores were 0.8082 for *Safe* prompts and 0.9512 for *Unsafe* prompts.

3.3 Late Fusion

The LinearSVC and E5 branches produced scores on different numerical scales. We therefore standardized the score from each branch using its development-set mean μ_b and standard deviation σ_b:

$$z_b(s) = \frac{s - \mu_b}{\sigma_b}. \tag{2}$$

The standardized scores were combined through late fusion:

$$s_{\text{hybrid}} = (1 - \lambda) z_{\text{SVC}} + \lambda z_{\text{E5}}, \tag{3}$$

where λ controls the contribution of the semantic branch. We evaluated E5 weights from 0.0 to 1.0 in increments of 0.1. The best development result was obtained with $\lambda = 0.20$, assigning weights of 0.80 and 0.20 to the LinearSVC and E5 branches, respectively. The resulting hybrid system achieved a development Macro-F1 of 0.9346. For the final submission, we also constructed a lookup table from normalized training and development prompts whose observed labels were unambiguous. When a test prompt exactly matched an entry in this table, the stored label replaced the model prediction. This exact-match recovery rule applied to 31 blind-test prompts. Because blind-test labels were unavailable during prediction, the individual contribution of this recovery rule could not be evaluated separately.

System	Macro-F1	Accuracy	E5 Weight
Frozen E5 + LR	0.880	0.922	1.000
LinearSVC	0.933	0.958	0.000
Hybrid	**0.935**	N/A	0.200

Table 1: Development results

4 Results

4.1 Development Results

Table 1 presents the results on the 2,405-example development set. LinearSVC was the strongest individual branch, achieving a Macro-F1 of 0.933 and an accuracy of 0.958. The frozen E5 branch achieved a lower Macro-F1 of 0.880 and an accuracy of 0.922. The hybrid system achieved the highest development Macro-F1, 0.935, representing an absolute improvement of approximately 0.002 over LinearSVC alone. The small magnitude of this improvement is important. It shows that most of the predictive performance came from the lexical branch and that the frozen semantic representation supplied only limited complementary information. The result should therefore not be interpreted as evidence that the two branches contributed equally or that frozen E5 embeddings were competitive with the task-specific lexical representation.

4.2 Fusion Ablation

Table 2 and Figure 2 provide an ablation over the E5 fusion weight. A weight of 0.0 represents LinearSVC alone, whereas a weight of 1.0 represents the frozen E5 branch alone. Performance increased from 0.933 to 0.935 when the E5 weight increased

E5 fusion weight	Development Macro-F1
0.000	0.933
0.100	0.933
0.200	0.935
0.300	0.934
0.400	0.931
0.500	0.931
0.600	0.929
0.700	0.920
0.800	0.907
0.900	0.895
1.000	0.880

Table 2: Development Macro-F1 across E5 fusion weights.

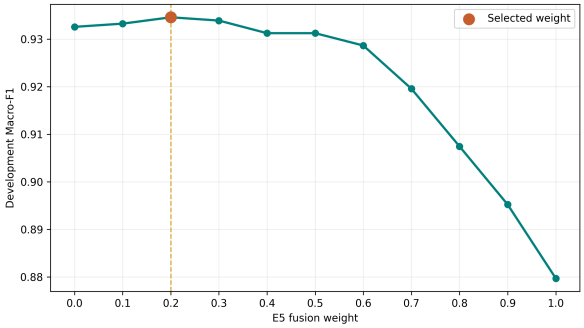


Figure 2: Development Macro-F1 across E5 fusion weights

from 0.0 to 0.2, but it declined as the semantic branch received more influence. The decline was gradual at first: Macro-F1 was 0.934 at an E5 weight of 0.3 and 0.931 at weights of 0.4 and 0.5. It became more pronounced at larger weights, falling to 0.907 at 0.8 and 0.880 when E5 was used alone. This pattern provides stronger evidence than the selected fusion result alone: frozen E5 features were useful only as a low-weight complement and could not replace the lexical representation. It also indicates that choosing equal branch weights would have produced a weaker system.

4.3 Class-Level Analysis

Table 3 shows that both individual branches performed substantially better on *Unsafe* prompts than on *Safe* prompts. For LinearSVC, the class-specific F1 scores were 0.8915 for *Safe* and 0.9736 for *Unsafe*. The disparity was larger for frozen E5, which achieved F1 scores of 0.8082 and 0.9512 for the two classes, respectively. For LinearSVC, safe-class recall of 0.882 was lower than its safe-class precision of 0.901. This indicates that the model more frequently mislabeled safe prompts as unsafe than it mislabeled unsafe prompts as safe. The E5 branch showed a related weakness: although its

Branch	Safe P	Safe R	Safe F1	Unsafe F1
LinearSVC	0.901	0.882	0.892	0.974
Frozen E5 + LR	0.788	0.830	0.808	0.951

Table 3: Development class-level results for the individual branches.

Metric	Score
Accuracy	**0.780**
Macro precision	**0.833**
Macro recall	**0.700**
Macro-F1	**0.715**

Table 4: Official test results for the submitted hybrid system.

safe-class recall was 0.830, its safe-class precision was only 0.788, indicating more confusion between the classes than in the lexical branch. These findings suggest that future improvements should focus on context-sensitive distinctions between genuinely harmful requests and safe prompts that contain harm-related terminology.

4.4 Test Results

On the test set, the submitted hybrid system achieved an accuracy of 0.780, macro precision of 0.833, macro recall of 0.700, and Macro-F1 of 0.715, as shown in Table 4. Macro precision was considerably higher than macro recall, indicating that the system's principal weakness was failing to recover examples consistently across both classes rather than making uniformly imprecise predictions. The test Macro-F1 was 0.220 lower than the development Macro-F1. This large difference shows that the development results substantially overestimated blind-test performance. One plausible explanation is that sparse lexical features benefited from wording and character patterns represented in the development split but generalized less effectively to prompts expressed using different constructions. However, because the blind-test labels and per-example errors were not available for post-submission inspection, the precise contributions of lexical shift, dialect distribution, label distribution, and threshold selection cannot be separated.

5 Conclusion

We presented a lightweight hybrid system for ArGuard 2026 Subtask B1 that combined normalized Arabic word- and character-level TF-IDF features with frozen `multilingual-e5-small` em-

beddings. LinearSVC was the strongest individual branch, while the semantic branch provided a small benefit when assigned a fusion weight of 0.20. The resulting system achieved a development Macro-F1 of 0.935, compared with 0.933 for LinearSVC and 0.880 for E5 alone. The fusion ablation showed that larger E5 contributions consistently reduced performance, establishing the frozen encoder as a limited complementary component rather than a replacement for lexical modeling. On the test set, the system achieved 0.780 accuracy, 0.833 macro precision, 0.700 macro recall, and 0.715 Macro-F1.

6 Ethical Considerations

Prompt-safety classification involves sensitive topics such as violence, harassment, hate speech, self-harm, sexual content, and fraud. Researchers may be exposed to harmful language during dataset review and model development. We used the task data only for research and did not reproduce harmful prompts in this paper.

7 Limitations

The fusion weight, score scaling, and model settings were selected using one development split, which may have led to overly high development results. The E5 encoder was frozen, its input was limited to 128 tokens, and the tested logistic-regression range ended at the best boundary value of $C = 4.0$. Because blind-test labels were not available, we could not analyze individual errors, dialects, or harm categories. Therefore, the results should not be generalized to sparse models or multilingual encoders in general.

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